

SUSHANT KUMAR

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SUMMARY

Engineering graduate with hands-on experience in AI-based computer vision, electronic instrumentation, testing, and medical equipment QA. Currently working at IRDE-DRDO on AI projects involving computer vision and automated detection. Proficient in Python, C/C++, MATLAB, OpenCV, TensorFlow, and Jetson Nano. Experienced in real-time image processing, signal analysis, fault diagnosis, calibration, and embedded systems.

EDUCATION

B.Tech in Biomedical Engineering North Eastern Hill University, Shillong, Meghalaya	2021 – 2025 CGPA: 6.59
Higher Secondary (PCM) Langat Singh College, Muzaffarpur, Bihar	2020 61.4%
Secondary Shree Baldev High School, Keotsa Baruari, Bihar	2018 53.8%

EXPERIENCE

QA Engineer – AI Department IRDE-DRDO, Dehradun	Nov 2025 – Present
- Working on AI-based computer vision and automated detection projects in the AI department. - Supporting QA activities: data validation, pipeline testing, output verification, and documentation. - Improving model reliability and performance through systematic testing and cross-functional collaboration.	
AI Intern IRDE-DRDO, Dehradun	Jan 2025 – Feb 2025
- Developed real-time cancer cell detection pipeline on Jetson Nano using IR thermal camera and Python. - Applied Laplacian edge detection, CLAHE, and Gaussian Blur for noise reduction and contrast enhancement. - Optimized image processing enabling clear tumour boundary detection in IR and MRI scans. - Contributed to algorithm design and parameter tuning, improving detection accuracy.	
Medical Equipment Intern Instrument Care Services, Kolkata	Jan 2023 – Feb 2023
- Studied working principles, block diagrams, and electronic signal flow of ECG machines, defibrillators, ICU ventilators, and dialysis units; analysed biosensor and transducer integration in life-critical devices. - Performed hands-on fault diagnosis, calibration, and preventive maintenance on multiparameter monitors, pacemakers, and syringe pumps using oscilloscopes and electronic test equipment. - Gained insight into AI-assisted patient monitoring trends and sensor-based early-warning systems used in modern ICU environments. - Observed signal-conditioning circuits for SpO2, ECG, and real-time vitals tracking — bridging electronic instrumentation with AI-driven biomedical applications.	

PROJECTS

Real-Time Cancer Cell Detection System Python, OpenCV, Jetson Nano, CLAHE, Laplacian Edge Detection
Built IR thermal + MRI based pipeline for tumour boundary detection using image enhancement techniques on Jetson Nano.
Bio-Sensor for BUN & Creatinine Detection Quantum Dots, Fluorescence Spectroscopy, Point-of-Care
- Synthesizing and conjugating quantum dots with biomolecules for fluorescence detection of BUN and creatinine. - Designing and fabricating a portable point-of-care device, with lab validation in progress.
Anchoring Mechanism for TMVR Device Biomedical Design, CAD, Design Thinking, IPR
- Designed anchoring mechanism for Transcatheter Mitral Valve Replacement (TMVR) studying heart function and device principles. - Applied design thinking at Impact Week and attended IPR workshop for biomedical innovation.

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB
AI/ML Frameworks: TensorFlow, OpenCV, VGG, NumPy, Pandas
Embedded Systems: Jetson Nano, 3D Printing, Autodesk 360
Medical Devices: ECG, Defibrillator, ICU Ventilator, Multiparameter Monitor
Domain Expertise: Signal Processing, MRI, Sensors & Transducers, Bioelectronics, QA & Testing

ACHIEVEMENTS & CERTIFICATIONS

- 2nd Position – Biocraft Competition, IIT Kharagpur.
- Satellite Communication & Navigation Training – NESAC, Dept. of Space, Govt. of India (2024).
- Yuvamanthan Model United Nations (YUMN) – Delegate, policy discussions & public speaking (2024).
- International Conference on Neuroscience & Neurological Disorders – SNCI, NEHU, Shillong (2023).
- Impact Week Meghalaya – Design Thinking Challenge for urban traffic solutions (2023).